

Regulation of skeletal muscle mass in mice by a new TGF-beta superfamily member.

The transforming growth factor-beta (TGF-beta) superfamily encompasses a large group of growth and differentiation factors playing important roles in regulating embryonic development and in maintaining tissue homeostasis in adult animals. Using degenerate polymerase chain reaction, we have identified a new murine TGF-beta family member, growth/differentiation factor-8 (GDF-8), which is expressed specifically in developing and adult skeletal muscle. During early stages of embryogenesis, GDF-8 expression is restricted to the myotome compartment of developing somites. At later stages and in adult animals, GDF-8 is expressed in many different muscles throughout the body. To determine the biological function of GDF-8, we disrupted the GDF-8 gene by gene targeting in mice. GDF-8 null animals are significantly larger than wild-type animals and show a large and widespread increase in skeletal muscle mass. Individual muscles of mutant animals weigh 2-3 times more than those of wild-type animals, and the increase in mass appears to result from a combination of muscle cell hyperplasia and hypertrophy. These results suggest that GDF-8 functions specifically as a negative regulator of skeletal muscle growth.